Why the SIR Model Maps onto Bond Percolation

(and why the SI model does not)

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in response to a question from Sang Hoon Lee

Abstract

A common first guess is that the SI model, in which infected nodes stay infected forever and the infected cluster only grows, should be the epidemic counterpart of percolation. In fact it is the SIR model whose *final outbreak* is a bond-percolation cluster. Percolation is a static question of which edges are *open* and which are *closed*. Recovery in SIR is exactly what allows an edge to close permanently; without recovery (SI) every edge eventually transmits, and the only remaining randomness is *when* nodes are reached, which is first-passage percolation rather than ordinary percolation.

1 Each edge gets exactly one chance

Consider an infected node i and a susceptible neighbour j. Infection crosses the edge $i \to j$ as a Poisson process of rate β, so the first attempt occurs after an exponentially distributed waiting time c_{ij}. In the SIR model node i is infectious only for a time τ, after which it is removed permanently. The edge therefore transmits if and only if $c_{ij} < \tau$ (Fig. 1). For a fixed infectious period this happens with the *transmissibility*

$$T = \Pr(c_{ij} < \tau) = 1 - e^{-\beta\tau}. \tag{1}$$

If the clock has not rung by time τ, the edge has *failed for good*: i will never be infectious again. This is precisely the closed bond of percolation.

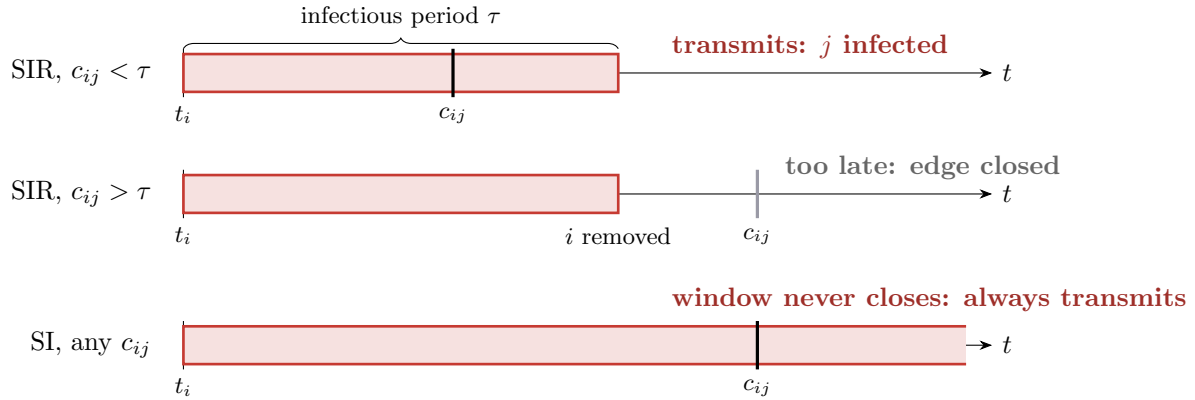


Figure 1: The edge $i \to j$ as a race between the transmission clock $c_{ij} \sim \mathrm{Exp}(\beta)$, measured from i's own infection time t_i, and the end of i's infectious window (shaded). In SIR the window closes, so the edge succeeds with probability $T < 1$ and otherwise fails permanently. In SI the window never closes and every edge eventually succeeds.

Two facts make this a percolation problem.

1. **Independence.** For a fixed τ, the clocks on different edges are independent, so each edge succeeds independently with probability T.

2. **Time drops out.** The clock c_{ij} is measured from t_i, whenever that happens to be. Whether j is *eventually* infected depends only on whether some chain of successful edges connects it to the seed, not on the timing. If j was already infected by someone else, the history changes but the final set does not.

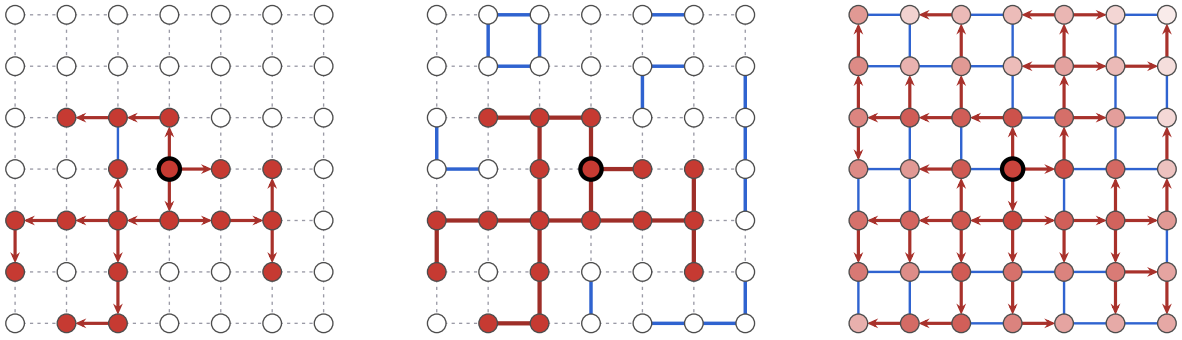
One can therefore draw all the clocks in advance, declare each edge open or closed, and read off the outbreak as a cluster:

$$\boxed{\text{final SIR outbreak from a seed } = \text{ bond-percolation cluster of the seed, with } p = T.} \quad (2)$$

Strictly, each edge has two clocks, one per direction, but the exploration only ever queries the direction leading away from the endpoint infected first. Each edge is therefore examined at most once, and the set reached has the same distribution as an ordinary undirected bond-percolation cluster.

2 The mapping on a small lattice

Figure 2 uses a single set of clocks on a 7×7 square lattice with $\tau = 1$ and $T = 0.5$. Panel (a) is an event-driven SIR simulation; panel (b) simply marks each edge open if its clock beat τ, and the seed's cluster coincides node for node with the infected set of panel (a). Panel (c) feeds *the same clocks* into an SI simulation: nothing ever closes, so the whole lattice is infected and only the arrival times carry information.



(a) SIR dynamics: infected nodes (red) and the transmission tree (arrows).

(b) Bond percolation, open with prob. T: the seed's cluster is the SIR outbreak.

(c) SI with the same clocks: everyone is infected; shading shows arrival time (darker is earlier).

Figure 2: One realisation of the clocks, three readings. Solid edges are open ($c < \tau$) and dotted grey edges are closed; in (b) the open edges of the seed's cluster are drawn thick and red. The seed has a thick outline. In (a) and (b) the infected set and the open cluster are identical (18 nodes). In (c) the arrival time of each node is its shortest-path distance from the seed, with edge lengths c_{ij}; this is first-passage percolation.

3 Transmissibility, and why SI is trivial

Equation (1) shows that the SIR control parameters (β, τ) enter the final outbreak only through the single combination T, which plays the role of the bond-occupation probability p. In SI, $\tau = \infty$ and $T = 1$ for every $\beta > 0$ (Fig. 3): percolation sits at $p = 1$, every node in the seed's connected component is infected, and there is no threshold.

If the infectious period is random, say exponential with recovery rate γ as in the standard Markovian SIR model, the average transmissibility becomes $T = \beta/(\beta + \gamma)$. However, all edges leaving node i share the same τ_i: a node that recovers early fails on all its edges together. The occupation variables are then correlated, and the exact mapping is to a semi-directed, correlated

percolation. Final outbreak *sizes* still follow from the percolation picture with this T, but the *probability* that a large outbreak occurs differs from the cluster-size prediction.

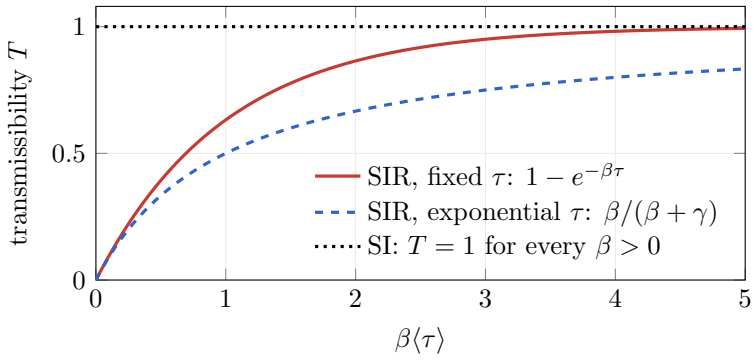


Figure 3: Transmissibility as a function of the mean infectious period. Any finite infectious period gives $T < 1$, and hence a nontrivial percolation problem; the SI limit pins T to 1.

4 A quantitative check

On a random network the percolation picture predicts the epidemic threshold and the outbreak size directly. For an Erdős–Rényi network with mean degree $\langle k\rangle$, the threshold is $T_c = \langle k\rangle/(\langle k^2\rangle - \langle k\rangle) = 1/\langle k\rangle$, and the giant-cluster fraction S solves

$$S = 1 - e^{-\langle k\rangle TS}. \tag{3}$$

With a fixed infectious period the edges are independent, so both the size of a large outbreak and the probability that one occurs should equal S. Figure 4 compares this prediction with direct event-driven SIR simulations.

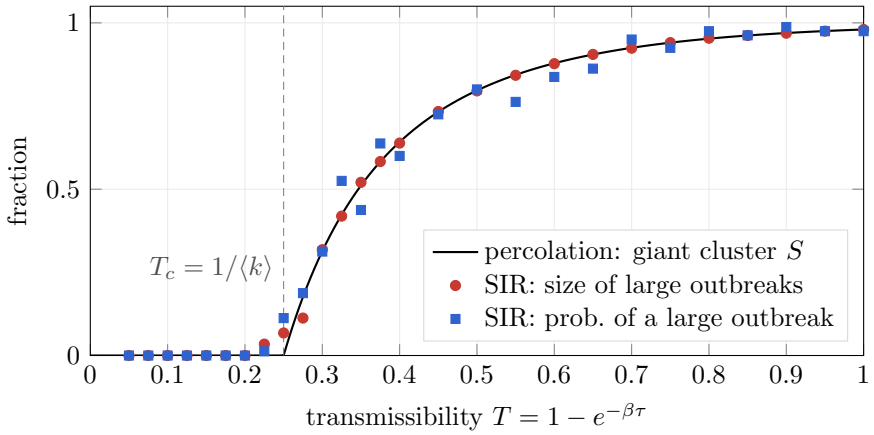


Figure 4: SIR on Erdős–Rényi networks ($N = 4000$, $\langle k\rangle = 4$, fixed $\tau = 1$, 80 runs per point, a "large" outbreak being one that infects more than 2% of the nodes). Both the outbreak size and the outbreak probability follow the giant-cluster curve of bond percolation. Deviations near T_c are finite-size effects.

5 Summary

Model	What is random	Percolation counterpart	Threshold
SIR	*whether* each edge transmits	bond percolation, $p = T$	T_c (nontrivial)
SI	only *when* each edge transmits	first-passage percolation	none ($T = 1$)
SIS	reinfection, ongoing in time	contact process / directed percolation	λ_c (dynamic)

Percolation needs edges that can *permanently fail*. Recovery in SIR provides exactly that, and once each edge has its single, independent chance, time disappears from the question of who is eventually infected. SI never lets an edge fail; it only delays it. Its natural percolation-type description is therefore first-passage percolation, which concerns *when* nodes are reached rather than *whether*. SIS keeps time essential through reinfection and belongs to the directed-percolation class rather than to static percolation.

Credits

The question, why the SIR model rather than the SI model corresponds to simple percolation, was posed by Sang Hoon Lee. The explanation, figures, simulations and this note were written by Claude Opus 5.5 (Anthropic) in conversation with him.

Further reading

- P. Grassberger, "On the critical behavior of the general epidemic process and dynamical percolation," *Math. Biosci.* **63**, 157 (1983).
- M. E. J. Newman, "Spread of epidemic disease on networks," *Phys. Rev. E* **66**, 016128 (2002).
- E. Kenah and J. M. Robins, "Second look at the spread of epidemics on networks," *Phys. Rev. E* **76**, 036113 (2007).
- J. M. Hammersley and D. J. A. Welsh, "First-passage percolation, subadditive processes, stochastic networks, and generalized renewal theory," in *Bernoulli, Bayes, Laplace Anniversary Volume* (Springer, 1965).